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OFF-AXIS THROUGH THE LENS MUTUALLY COHERENT DARK FIELD IMAGING SYSTEM WITH INCOHERENT LIGHT FOR OVERLAY METROLOGY

Abstract

A metrology system is disclosed. The system includes an incoherent illumination source to generate incoherent illumination. The system includes a diffraction grating to split the incoherent illumination. The system includes an objective lens to direct one or more pairs of mutually coherent illumination beams to a metrology target on a sample and collect sample light associated with diffraction of pairs of mutually coherent illumination beams. The system includes a mask configured to pass a single nonzero-order diffraction beam and block a zero-order diffraction beam associated with each of the mutually coherent illumination beams. The system includes a detector configured to generate an image of the metrology target based on light passed by the mask. The system includes a controller communicatively coupled to the detector, including one or more processors configured to generate one or more metrology measurements of the sample based on the image.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Application Ser. No. 63/556,863, filed Feb. 22, 2024, entitled OFF-AXIS THROUGH THE LENS MUTUALLY-COHERENT DARK FIELD IMAGING SYSTEM WITH INCOHERENT LIGHT FOR OVERLAY METROLOGY, which is incorporated herein by reference in the entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to overlay metrology and, more particularly, to dark-field imaging overlay metrology.

BACKGROUND

[0003] Overlay metrology systems typically characterize the overlay alignment of multiple layers of a sample by measuring the relative positions of metrology target features located on layers of interest. As the size of fabricated features decreases and the feature density increases, the demands on overlay metrology systems needed to characterize these features increase. In particular, smaller features require more sensitive and more accurate measurements of small alignment errors. Accordingly, it is desirable to develop systems and methods to address these demands.

SUMMARY

[0004] A metrology system is disclosed, in accordance with one or more embodiments of the present disclosure. In embodiments, the metrology system includes an incoherent illumination source to generate incoherent illumination. In embodiments, the metrology system includes a diffraction grating to split the incoherent illumination into one or more pairs of mutually coherent illumination beams. In embodiments, the metrology system includes an objective lens to direct the one or more pairs of mutually coherent illumination beams to a metrology target on a sample, wherein the metrology target is configured in accordance with a metrology recipe to include periodic features associated with two or more lithographic exposures, wherein the objective lens further collects sample light associated with diffraction of the one or more pairs of mutually coherent illumination beams. In embodiments, the metrology system includes one or more masks, wherein the one or more masks are configured in accordance with the metrology recipe to pass a single nonzero-order diffraction beam and block a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams. In embodiments, the metrology system includes a detector configured to generate an image of the metrology target based on light passed by the one or more masks. In embodiments, the metrology system includes a controller communicatively coupled to the detector, wherein the controller includes one or more processors configured to execute program instructions causing the one or more processors to generate one or more metrology measurements of the sample based on the image in accordance with the metrology recipe.

[0005] A metrology method is disclosed, in accordance with one or more embodiments of the present disclosure. In embodiments, the metrology method includes generating, with an incoherent illumination source, incoherent illumination. In embodiments, the metrology method includes diffracting, with a diffraction grating, the incoherent illumination into one or more pairs of mutually coherent illumination beams. In embodiments, the metrology method includes directing, with an objective lens, the one or more pairs of mutually coherent illumination beams to a metrology target on a sample, wherein the metrology target is configured in accordance with a metrology recipe to include periodic features associated with two or more lithographic exposures. In embodiments, the metrology method includes collecting, with the objective lens, sample light associated with diffraction of the one or more pairs of mutually coherent illumination beams. In embodiments, the metrology method includes passing, with one or more masks, a single nonzero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams. In embodiments, the metrology method includes blocking, with the one or more masks, a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams; generating, with a detector, an image of the metrology target based on light passed by the one or more masks. In embodiments, the metrology method includes generating one or more metrology measurements of the sample based on the image in accordance with the metrology recipe.

[0006] A metrology system is disclosed, in accordance with one or more embodiments of the present disclosure. In embodiments, the metrology system includes a diffraction grating including one or more diffraction grating cells having a direction of periodicity configured to split incoherent illumination from an incoherent illumination source into one or more pairs of mutually coherent illumination beams. In embodiments, the metrology system an objective lens to direct the one or more pairs of mutually coherent illumination beams from the beamsplitter to a metrology target on a sample, wherein the metrology target is configured in accordance with a metrology recipe including two or more lithographic exposures to generate periodic features having a direction of periodicity arranged into one or more target cells, wherein the direction of periodicity of a target cell corresponds to the direction of periodicity of a diffraction grating cell, wherein the objective lens further collects sample light associated with diffraction of the one or more pairs of mutually coherent illumination beams. In embodiments, the metrology system one or more masks, wherein the one or more masks are configured in accordance with the metrology recipe to pass a single nonzero-order diffraction beam and block a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams. In embodiments, the metrology system a detector configured to generate an image of the metrology target based on light passed by the one or more masks. In embodiments, the metrology system a controller communicatively coupled to the detector, wherein the controller includes one or more processors configured to execute program instructions causing the one or more processors to generate one or more metrology measurements of the sample based on the image in accordance with the metrology recipe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The detailed description is described with reference to the accompanying figures. The use of the same reference numbers in different instances in the description and the figures may indicate similar or identical items. Various embodiments or examples ("examples") of the present disclosure are disclosed in the following detailed description and the accompanying drawings. The drawings are not necessarily to scale. In general, operations of disclosed processes may be performed in an arbitrary order, unless otherwise provided in the claims. In the drawings:

[0008] FIG. **1**A illustrates a block diagram illustrating an overlay metrology system, in accordance with one or more embodiments of the present disclosure.

[0009] FIG. **1**B illustrates a simplified schematic view of an overlay metrology sub-system, in accordance with one or more embodiments of the present disclosure.
[0010] FIG. **2**A illustrates a conceptual diagram of illuminating a metrology target with a pair of mutually coherent illumination beams, in accordance with one or more embodiments of the present disclosure.
[0011] FIG. **2**B illustrates a top view of an illumination pupil with a single pair of mutually coherent illumination beams, in accordance with one or more embodiments of the present disclosure.
[0012] FIG. **2**C illustrates a top view of an illumination pupil with two pairs of mutually coherent illumination beams, in accordance with one or more embodiments of the present disclosure.
[0013] FIG. **2**D illustrates a top view of a collection pupil with a mask for the single pair of mutually coherent illumination beams of FIG. **2**B, in accordance with one or more embodiments of the present disclosure.
[0014] FIG. **2**E illustrates a top view of a collection pupil with a mask for the two pairs of mutually coherent illumination beams of FIG. **20**, in accordance with one or more embodiments of the present disclosure.
[0015] FIG. **2**F illustrates a top view of a collection pupil with a mask for the single pair of mutually coherent illumination beams of FIG. **2**B configured to pass second order diffraction, in accordance with one or more embodiments of the present disclosure.
[0016] FIG. **3**A illustrates a diffraction grating, in accordance with one or more embodiments of the present disclosure.
[0017] FIG. **3**B illustrates an overlay target, in accordance with one or more embodiments of the present disclosure.
[0018] FIG. **4** illustrates a flow diagram illustrating a method, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0019] Before explaining one or more embodiments of the disclosure in detail, it is to be understood that the embodiments are not limited in their application to the details of construction and the arrangement of the components or steps or methodologies set forth in the following description or illustrated in the drawings. In the following detailed description of embodiments, numerous specific details may be set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art having the benefit of the instant disclosure that the embodiments disclosed herein may be practiced without some of these specific details. In other instances, well-known features may not be described in detail to avoid unnecessarily complicating the instant disclosure.
[0020] Embodiments of the present disclosure are directed to systems and methods for overlay metrology with pairs of mutually coherent illumination beams (e.g., mutually coherent illumination beam pairs). The pairs of mutually coherent illumination beams may originate from an incoherent illumination source, where the illumination from the incoherent illumination source is diffracted by a diffraction grating to form one or more pairs of mutually coherent illumination beams.
[0021] This may have the advantage of suppressing speckle in an image and reducing edge effect. Additionally, formed dark field images may be color corrected so long as collected illumination is corrected, which may result in a formed grating image being independent of an illumination source wavelength, which may allow a larger bandwidth of light to be used.
[0022] Overlay metrology using pairs of mutually coherent illumination beams generated by a coherent light source is disclosed in U.S. Pat. No. 12,032,300, entitled “Imaging overlay with mutually-coherent oblique illumination” and issued on Jul. 9, 2024, and U.S. patent application Ser. No. 18/742,869, entitled “Imaging overlay with mutually-coherent oblique illumination” and filed on Jun. 13, 2024, which are herein incorporated by reference in its entirety.
[0023] A metrology target may be illuminated with a pair of mutually-coherent illumination beams

at symmetric azimuth angles for each measurement direction of interest. For example, an overlay measurement for a single measurement direction may utilize one pair of mutually-coherent illumination beams, while an overlay measurement for two measurement directions (e.g., two orthogonal measurement directions) may utilize two pairs of mutually-coherent illumination beams. Further, although illumination beams within each pair are mutually-coherent, it is not necessary for beams in different pairs to be mutually-coherent. Rather, it may be beneficial but not required that the pairs of illumination beams are incoherent with respect to each other.

[0024] A metrology target and/or an overlay metrology tool suitable for characterizing the metrology target may be configured according to a metrology recipe suitable for generating overlay measurements based on a desired technique. More generally, an overlay metrology tool may be configurable according to a variety of metrology recipes to perform overlay measurements using a variety of techniques and/or perform overlay measurements on a variety of metrology targets with different designs.

[0025] For example, a metrology recipe may include various aspects of a metrology target or a design of a metrology target including, but not limited to, a layout of target features on one or more sample layers, feature sizes, or feature pitches. As another example, a metrology recipe may include illumination parameters such as, but not limited to, an illumination wavelength, an illumination pupil distribution (e.g., a distribution of illumination angles and associated intensities of illumination at those angles), a polarization of incident illumination, a spatial distribution of illumination, or a sample height. By way of another example, a recipe of an overlay metrology tool may include collection parameters such as, but not limited to, a collection pupil distribution (e.g., a desired distribution of angular light from the sample to be used for a measurement and associated filtered intensities at those angles), collection field stop settings to select portions of the sample of interest, polarization of collected light, or wavelength filters.

[0026] In some embodiments, an overlay metrology tool is configured (e.g., according to a metrology recipe) to image a metrology target having periodic structures using a single non-zero diffraction lobe from each illumination beam in a mutually-coherent illumination beam pair. In this configuration, the periodic structures from the metrology target may be imaged in a dark-field imaging mode as sinusoidal interference patterns. Further, overlay may be determined by comparing relative phases of various imaged sinusoidal patterns of a metrology target. An overlay metrology system may be configured in multiple ways in accordance with the systems and method disclosed herein. In some embodiments, an overlay metrology system is configured to direct a pair of mutually-coherent illumination beams to a metrology target within a numerical aperture (NA) of an objective lens used to collected light from the metrology target for imaging, which is referred to herein as through-the-lens (TTL) illumination. In this configuration, the overlay metrology system may include one or more elements to block zero-order diffraction such that it does not contribute to image formation.

[0027] It is contemplated herein that overlay metrology based on mutually-coherent illumination beam pairs as disclosed herein may provide multiple advantages relative to existing image-based overlay metrology techniques based on incoherent illumination including, but not limited to, support of fine grating pitches, high image contrast, high image brightness, matched brightness between cells of a metrology target, insensitivity to monochromatic aberrations (e.g., defocus, or the like), minimal encroachment of cell edges, and/or minimal stray light.

[0028] For example, image contrast of periodic features is high (maximized in some cases) by interfering only two diffracted orders with equal amplitudes. As another example in the case of an r-AIM target, image brightness is matched between target cells since only grating pitches differ between cells.

[0029] In some embodiments, the overlay metrology measurements are used to generate correctables to control one or more additional process tools such as, but not limited to, a lithography tool, an etching tool, or a polishing tool.

[0030] FIG. **1**A illustrates a block diagram illustrating an overlay metrology system **100**, in accordance with one or more embodiments of the present disclosure.
[0031] In some embodiments, the overlay metrology system **100** includes an overlay metrology sub-system **102** configured to illuminate a metrology target **104** on a sample **106** with a pair of mutually-coherent illumination beams **108**. In particular, each of the illumination beams **108***a*,*b* may fully illuminate the entirety of a metrology target **104**. In this way, each cell of the metrology target **104** receives common illumination conditions to promote matched image brightness for all of the cells.
[0032] In embodiments, the sample **106** may be disposed on a sample stage (not shown) suitable for securing the sample **106** and further configured to position the metrology target **104** with respect to the illumination beams **108**.
[0033] The metrology target **104** may include various periodic features, which may have a periodicity. In embodiments, the metrology target **104** may have periodic features that have a periodicity along a single measurement direction (e.g., an x-direction or a y-direction). In embodiments, the metrology target **104** may include a first set of periodic features along a first measurement direction (e.g., an x-direction) and a second set of periodic features along a second measurement direction (e.g., a y-direction).
[0034] In embodiments, the periodic features of the metrology target may be arranged into two or more target cells.
[0035] It is contemplated herein that various metrology target designs are suitable for overlay measurements with mutually-coherent illumination beam pairs as disclosed herein. In embodiments, a metrology target **104** is an advanced imaging metrology (AIM) target. In this configuration, each cell of the metrology target **104** may include grating structures from different lithographic exposures in non-overlapping regions on one or more layers, where the grating structures from the different lithographic exposures have the same pitch. In embodiments, the metrology target **104** is a Moire target. In this configuration, each cell may include grating structures from different lithographic exposures in overlapping regions on two layers to form grating-over-grating structures or Moire structures, where the grating structures from the different lithographic exposures have different pitches. Further, a cell may include a pair of Moire structures in which the pitches on the constituent layers are reversed relative to each other. For example, a first
[0036] Moire structure may have a first pitch on a first layer and a second pitch on a second layer, while a second Moire structure may have the first pitch on the second layer and the second pitch on the first layer. Such a metrology target **104** may be referred to as a robust-AIM (r-AIM) metrology target and provides that an overlay measurement may be determined based on relative phases between the two Moiré structures.
[0037] The overlay metrology sub-system **102** may image a metrology target **104** based on a single non-zero diffraction lobe associated with each illumination beam **108** in each pair of mutually-coherent illumination beams **108**. In this way, the overlay metrology sub-system **102** may provide a dark-field image since zero-order diffraction of the illumination beams **108** does not contribute to image formation. Further, since each pair of illumination beams **108** is mutually-coherent, the single diffraction lobe associated with each illumination beam **108** in the pair interferes with its counterpart to form a sinusoidal interference pattern in the image. As a result, the various grating structures in the metrology target **104** may be imaged with high contrast as pure sinusoids such that overlay measurements may be generated based on comparisons of relative phases of the neighboring cell images in accordance with a metrology recipe.
[0038] In embodiments, the overlay metrology system **100** includes a controller **110** communicatively coupled to the overlay metrology sub-system **102**. The controller **110** may be configured to direct the overlay metrology sub-system **102** to generate dark-field images based on one or more selected metrology recipes. The controller **110** may be further configured to receive

data including, but not limited to, dark-field images from the overlay metrology sub-system **102**. Additionally, the controller **110** may be configured to determine overlay associated with a metrology target **104** based on the acquired dark-field images. As another example, the controller **110** may generate correctables to control, based on the overlay metrology measurements, one or more process tools such as, but not limited to, a lithography tool, an etching tool, or a polishing tool. Correctables may be generated to control one or more process tools in any combination of a feedback control loop or a feed-forward control loop. As an illustration, feedback correctables generated in response to metrology measurements on a sample **106** may control a process tool during the fabrication of additional samples in the same or different lots (e.g., in response to drifts of the process tools). As another illustration, feed-forward correctables generated in response metrology measurements on a sample **106** may be used to control a process tool during fabrication of additional features on the sample **106** in future process steps.

[0039] In some embodiments, the controller **110** includes one or more processors **112**. For example, the one or more processors **112** may be configured to execute a set of program instructions maintained in a memory **114**, or memory device.

[0040] The one or more processors **112** of a controller **110** may include any processor or processing element known in the art. For the purposes of the present disclosure, the term “processor” or “processing element” may be broadly defined to encompass any device having one or more processing or logic elements (e.g., one or more micro-processor devices, one or more application specific integrated circuit (ASIC) devices, one or more field programmable gate arrays (FPGAs), or one or more digital signal processors (DSPs)). In this sense, the one or more processors **112** may include any device configured to execute algorithms and/or instructions (e.g., program instructions stored in memory). In embodiments, the one or more processors **112** may be embodied as a desktop computer, mainframe computer system, workstation, image computer, parallel processor, networked computer, or any other computer system configured to execute a program configured to operate or operate in conjunction with the overlay metrology system **100**, as described throughout the present disclosure. Moreover, different subsystems of the overlay metrology system **100** may include a processor or logic elements suitable for carrying out at least a portion of the steps described in the present disclosure. Therefore, the above description should not be interpreted as a limitation on the embodiments of the present disclosure but merely as an illustration. Further, the steps described throughout the present disclosure may be carried out by a single controller or, alternatively, multiple controllers. Additionally, the controller **110** may include one or more controllers housed in a common housing or within multiple housings. In this way, any controller or combination of controllers may be separately packaged as a module suitable for integration into the overlay metrology system **100**.

[0041] The memory **114** may include any storage medium known in the art suitable for storing program instructions executable by the associated one or more processors **112**. For example, the memory **114** may include a non-transitory memory medium. By way of another example, the memory **114** may include, but is not limited to, a read-only memory (ROM), a random-access memory (RAM), a magnetic or optical memory device (e.g., disk), a magnetic tape, a solid-state drive, and the like. It is further noted that the memory **114** may be housed in a common controller housing with the one or more processors **112**. In some embodiments, the memory **114** may be located remotely with respect to the physical location of the one or more processors **112** and the controller **110**. For instance, the one or more processors **112** of the controller **110** may access a remote memory (e.g., server), accessible through a network (e.g., internet, intranet, and the like).

[0042] FIG. **1**B illustrates a simplified schematic view of an overlay metrology sub-system **102**, in accordance with one or more embodiments of the present disclosure.

[0043] In embodiments, the overlay metrology sub-system **102** includes an incoherent illumination source **116**. The incoherent illumination source **116** may be configured to generate incoherent illumination. The incoherent illumination source **116** may be any incoherent illumination source

116 known in the art suitable for generating incoherent illumination. For example, the incoherent illumination source **116** may be a lamp source (e.g. a laser-sustained plasma light source), a supercontinuum source, or a broadband illumination source (e.g., when the objective lens **124** is corrected for chromatic aberration and the image is color-corrected). By way of another example the incoherent illumination source **116** may be an incandescent light bulb (e.g., a traditional tungsten filament bulb), fluorescent lights, or most light emitting diode (LED) lights. A singular incoherent illumination source **116** may generate the entirety of the light used to illuminate the sample **106**.

[0044] In embodiments, the overlay metrology sub-system **102** includes a diffraction grating **118**. The diffraction grating **118** may be configured to split the incoherent illumination from the incoherent illumination source **116** into one or more pairs of mutually-coherent illumination beams **108***a,b*. The diffraction grating **118** may be located at a field plane conjugate to the sample **106**.

[0045] The diffraction grating **118** may be segmented to have different properties (e.g., pitch or orientation). This may permit different areas (e.g., different metrology targets **104**) on a sample **106** to be illuminated by independent off-axis illumination. Multiple diffraction gratings **118** may also be arranged on a slide or a wheel to provide additional configurations of off-axis illumination.

[0046] In embodiments, the overlay metrology sub-system **102** includes an illumination pupil **120**, wherein an illumination beam **108** passes through the illumination pupil **120**.

[0047] In embodiments, the overlay metrology sub-system **102** includes one or more illumination optics **134**. For example, each of the illumination optics **134** may include, but is not required to include, one or more illumination lenses (e.g., to control a spot size of the illumination beam **108** on the metrology target **104**, to relay pupil and/or field planes, or the like), one or more polarizers to adjust the polarization of the illumination beam **108**, one or more filters, one or more beam splitters, one or more diffusers, one or more homogenizers, one or more apodizers, one or more beam shapers, or one or more mirrors (e.g., static mirrors, translatable mirrors, scanning mirrors, or the like).

[0048] In embodiments, the overlay metrology sub-system **102** includes an objective lens **124**. The objective lens **124** may direct the one or more pairs of mutually-coherent illumination beams **108***a,b* to a metrology target **104** on a sample **106**. Additionally, the objective lens **124** may direct sample light **136** associated with the diffraction of the one or more pairs of mutually-coherent illumination beams **108***a,b* to a detector **132**.

[0049] In embodiments, the overlay metrology sub-system **102** includes a tube lens **130**. The tube lens **130** may be positioned near a detector **132** and be configured to direct light from the sample **106** to the detector **132**.

[0050] In embodiments, the overlay metrology sub-system **102** includes a detector **132**. The detector **132** may be configured to generate an image of the metrology target **104** based on light (e.g., sample light **136**). The detector **132** may be communicatively coupled to the controller **110**. The one or more processors **112** of the controller **110** may be configured to execute one or more sets of program instructions which may cause the one or more processors **112** to generate one or more metrology measurements of the sample **106** based on the image, in accordance with the metrology recipe.

[0051] Referring now to FIGS. **2**A-**2**C, aspects of illumination with the overlay metrology sub-system **102** are described in greater detail. FIG. **2**A generally illustrates illumination of the metrology target. FIG. **2**B illustrates an illumination pupil **120** for dipole illumination. FIG. **2**C illustrates an illumination pupil **120** for quadrupole illumination.

[0052] FIG. **2**A illustrates a conceptual diagram of illuminating a metrology target **104** with a pair of mutually-coherent illumination beams **108***a,b*, in accordance with one or more embodiments of the present disclosure.

[0053] FIG. **2**A depicts two illumination beams **108***a,b* as a pair of mutually-coherent illumination beams **108**. In some embodiments, the overlay metrology sub-system **102** directs a pair of mutually-coherent illumination beams **108** at symmetric incidence angles. For example, the illumination beams **108***a,b* have symmetric polar incidence angles (±α) and symmetric (e.g., opposing) azimuth incidence angles. In FIG. **2**A, this is illustrated by the two illumination beams **108***a,b* propagating in opposite azimuth directions in a plane of the figure.

[0054] FIG. **2**B illustrates a top view of an illumination pupil **120** with one pair of mutually-coherent illumination beams **108***a,b*, in accordance with one or more embodiments of the present disclosure. FIG. **2**B illustrates the pair of mutually-coherent illumination beams **108***a,b* as dipole illumination. In some embodiments, such a metrology target **104** includes grating structures along the X direction. In some embodiments, such a metrology target **104** includes grating structures along a different direction. In this case, the distribution in FIG. **2**B may be a rotated dipole configuration.

[0055] In embodiments, the one or more pairs of mutually-coherent illumination beams may be a single pair of mutually-coherent illumination beams **108***a,b*. The single pair of mutually-coherent illumination beams **108***a,b* may be used when periodic features on the metrology target **104** have a periodicity along a single measurement direction. Thus, the one or measurements generated from illuminating the metrology target **104** with a single pair of mutually-coherent illumination beams **108***a,b* may be associated with a single measurement direction (e.g., the measurement direction of periodicity). The single pair of mutually-coherent illumination beams **108***a,b* may be incident on the metrology target **104** at opposing azimuth angles aligned with the single measurement direction.

[0056] FIG. **2**C illustrates a top view of an illumination pupil **120** illustrating two pairs of mutually-coherent illumination beams **108***a,b,c,d*, in accordance with one or more embodiments of the present disclosure. In particular, FIG. **2**C depicts a first pair of mutually-coherent illumination beams **108***a,b* oriented along the X direction in the figure and a second pair of mutually-coherent illumination beams **108***c,d* oriented along the Y direction in the figure. In this way, FIG. **2**C illustrates two pairs of mutually-coherent illumination beams **108** as quadrupole illumination.

[0057] In embodiments, the one or more pairs of mutually-coherent illumination beams may be two pairs of mutually-coherent illumination beams **108***a,b,c,d* (e.g., a first pair of mutually-coherent illumination beams **108***a,b* and a second pair of mutually-coherent illumination beams **108***c,d*). It is contemplated herein that the quadrupole illumination distribution of FIG. **2**C may be suitable for imaging a two-dimensional (2D) metrology target **104** having grating structures oriented along two different (e.g., orthogonal) directions (e.g., a first measurement direction and a second measurement direction). In some embodiments, such a 2D metrology target **104** includes grating structures along the X and Y directions. In some embodiments, such a 2D metrology target **104** includes grating structures along two different orthogonal directions. In this case, the distribution in FIG. **2**C may be a rotated quadrupole configuration.

[0058] There may be a first pair of mutually-coherent illumination beams **108***a,b*. The first pair of mutually-coherent illumination beams **108***a,b* may be incident on the metrology target **104** at opposing azimuth angles aligned with the first measurement direction. The second pair of mutually-coherent illumination beams **108***c,d* may be incident on the metrology target at opposing azimuth angles aligned with the second measurement direction. Therefore, when the metrology target **104** includes two measurement directions, two pairs of mutually-coherent illumination beams **108***a,b,c,d* may be used to generate measurements for both measurement directions.

[0059] In some embodiments, all four illumination beams **108***a,b,c,d* are mutually-coherent. In some embodiments, different pairs of mutually-coherent illumination beams **108** are incoherent with respect to each other. For example, the first pair of mutually-coherent illumination beams **108***a,b* may be mutually-coherent with respect to each other, but incoherent with respect to the second pair of mutually-coherent illumination beams **108***c,d*.

[0060] Referring again to FIG. **1**B, collection of illumination from the sample **106** is described.

[0061] In embodiments, the overlay metrology sub-system **102** includes one or more optical elements to combine an illumination pathway with a collection pathway such that the objective lens **124** may both direct the mutually-coherent illumination beams **108**_a_, **108**_b_ to the metrology target **104** and collect associated diffracted light (e.g., sample light **136**). Put another way, such optical elements may enable TTL illumination and imaging with the objective lens **124**.

[0062] For example, as depicted in FIG. **1**B, the overlay metrology sub-system **102** may include a beamsplitter **122** to direct the mutually-coherent illumination beams **108**_a_, **108**_b_ to the objective lens **124** and pass diffracted light collected by the objective lens **124** towards the detector **132**.

[0063] As another example, the overlay metrology sub-system **102** may include a patterned mirror designed with a reflective pattern to direct the mutually-coherent illumination beams **108**_a_, **108**_b_ to the objective lens **124** and a transmissive pattern to pass diffracted light collected by the objective lens **124** towards the detector **132**. For instance, the patterned mirror may have an annular reflective pattern designed to direct the mutually-coherent illumination beams **108**_a_, **108**_b_ to the objective lens **124**, where the diffracted light collected by the objective lens **124** may pass through a transmissive portion on interior or exterior portions of the annular reflective pattern.

[0064] In embodiments, the overlay metrology sub-system **102** includes one or more masks **126** configured to pass a single, nonzero-order diffraction beam (e.g., a +1 order diffraction beam or a −1 order diffraction beam) associated with each of the one or more pairs of mutually-coherent illumination beams **108**_a,b_ while blocking a zero-order diffraction beam associated with each of the one or more pairs of mutually-coherent illumination beams **108**_a,b_. Therefore, a dark field image may be produced by the detector **132** because zero-order diffraction is blocked by the mask **126**.

[0065] The one or more masks **126** may include any component or combination of components suitable for passing pass a single diffraction beam associated with each of the mutually-coherent illumination beams **108**_a,b_ (e.g., a single first-order diffraction beam, a single second-order diffraction beam, or the like) while blocking zero-order diffraction beams associated with mutually-coherent illumination beams **108**_a,b_. Further, the one or more masks **126** may be placed at any suitable location in the overlay metrology sub-system **102**.

[0066] In embodiments, the overlay metrology sub-system **102** includes a mask **126** located at a collection pupil **128**. For example, a mask **126** located at a collection pupil **128** may include one or more apertures surrounded by opaque portions, where desired diffraction orders may pass through the one or more apertures of the mask **126** while the opaque portions may block the zero-order diffraction beams as well as any other undesired light. As an illustration, a mask **126** located at a collection pupil **128** may have an annular aperture to pass desired diffraction orders to the detector **132**.

[0067] In embodiments, if the overlay metrology sub-system **102** includes a patterned mirror designed with a reflective pattern to direct the mutually-coherent illumination beams **108**_a_, **108**_b_ to the objective lens **124** and a transmissive pattern to pass diffracted light collected by the objective lens **124** towards the detector **132** as described above, this patterned mirror may operate as a mask **126**.

[0068] In embodiments, the mask **126** may be configured as an annular mask with a central aperture. Therefore, the first-order diffraction used to create an image of the metrology target **104** may pass through an opening in the center of the mask **126** and the collection pupil **128**.

[0069] Referring now to FIGS. **2**D-F, masks **126** are described in greater detail. FIG. **2**D illustrates a mask **126** used with the dipole illumination of FIG. **2**B. FIG. **2**E illustrates a mask **126** used with the quadrupole illumination of FIG. **2**C. FIG. **2**F illustrates a mask **126** for imaging with second order diffraction.

[0070] FIG. **2**D illustrates a top view of a collection pupil **128** with a mask **126** for the single pair of mutually-coherent illumination beams **108**_a,b_ of FIG. **2**B, in accordance with one or more embodiments of the present disclosure.

[0071] In FIG. **2**D, the mask **126** on the collection pupil **128** allows only two diffracted beams of light through. The mask **126** allows a −1-order diffracted beam (e.g., associated with illumination beam **108***a*) and a +1-order diffracted beam (e.g., associated with illumination beam **108***b*). All other diffracted beams are either blocked by the mask **126** or are diffracted outside the collection pupil **128**. Therefore, only the +1-order diffracted beam and the −1-order diffracted beam pass through the mask **126** and collection pupil **128** to the detector **132**. This results in dark field image of the metrology target **104** to be formed, as no 0-order diffracted beam reaches the detector **132**.

[0072] FIG. **2**E illustrates a top view of a collection pupil **128** with a mask **126** for the two pairs of mutually-coherent illumination beams **108***a,b,c,d* of FIG. **2**C, in accordance with one or more embodiments of the present disclosure.

[0073] In FIG. **2**E, the mask **126** on the collection pupil **128** allows only four diffracted beams of light through. The mask **126** allows a −1-order diffracted beam (e.g., associated with illumination beams **108***a* and **108***c*) and a +1-order diffracted beam (e.g., associated with illumination beams **108***b* and **108***d*). All other diffracted beams are either blocked by the mask **126** or are diffracted outside the collection pupil **128**. Therefore, only two +1-order diffracted beams and two −1-order diffracted beams pass through the mask **126** and collection pupil **128** to the detector **132**. This results in dark field image of the metrology target **104** to be formed, as no 0-order diffracted beam reaches the detector **132**.

[0074] While FIGS. **2**D and **2**E have been shown and described with relation to first-order diffraction, it should be noted that this is for illustrative purposes and not limiting purposes. Any nonzero-order diffraction may be used to create a dark field image for the sample **106**.

[0075] FIG. **2**F illustrates a top view of a collection pupil **128** with a mask **126** for the single pair of mutually-coherent illumination beams of FIG. **2**B configured to pass second order diffraction, in accordance with one or more embodiments of the present disclosure. For example, **2**F illustrates a collection pupil **128** to pass higher, non-zero order diffraction to the detector **132**. Therefore, the collection pupil may include a central obscuration to block first order diffraction. Combined with the mask **126**, the collection pupil of **128** may only allow second order diffraction to pass.

[0076] Additionally, the central obscuration **202** as shown in FIG. **2**F may be used in instances even when it is not the goal to pass second order diffraction to the detector **132**. For example, a diffraction grating **118** may not perfectly diffract the illumination beams **108**. Therefore, a central obscuration **202** may block unwanted diffraction based on an imperfect diffraction grating **118**.

[0077] As discussed above, it is contemplated that the patterned mirror may operate as a mask **126**. However, the patterned mirror may still work in conjunction with another mask **126**. In this way, the patterned mirror may block some diffraction from reaching the mask **126**, while the mask **126** blocks all other unwanted diffraction from reaching the detector **132**. In this way, the overlay metrology sub-system **102** effectively has two masks **126**.

[0078] FIG. **3**A illustrates a diffraction grating **118**, in accordance with one or more embodiments of the present disclosure. FIG. **3**B illustrates an overlay target **104**, in accordance with one or more embodiments of the present disclosure.

[0079] The diffraction grating **118** may be configured to generate one or more pairs of mutually-coherent illumination beams **108** by splitting incoherent illumination produced by an incoherent illumination source **116**. For example, when the metrology target **104** includes periodicity in a single measurement direction, the diffraction grating **118** may be configured to generate a single pair of mutually-coherent illumination beams **108** from incoherent illumination produced by the incoherent illumination source **116**.

[0080] Where the periodic features of the metrology target **104** are arranged into two or more target cells, the diffraction grating **118** may include periodic features arranged into two or more diffraction grating cells with a common arrangement as the two or more target cells. The number of diffraction grating cells **118***a,b,c,d* may be the same as the number of target cells. Additionally, the directions of periodicity of the two or more target cells may correspond to directions of periodicity of the periodic features in the two or more target cells **104***a,b,c,d*.

[0081] For example, in FIG. **3**B, the metrology target **104** includes periodic features arranged into

four target cells **104***a,b,c,d*. To correspond to the four target cells **104***a,b,c,d* on the metrology target **104**, the diffraction grating **118** in FIG. **3**A includes four diffraction grating cells **118***a,b,c,d*.
[0082] Additionally, the periodic features in the metrology target **104** in FIG. **3**B are arranged in two directions. Target cells **104***a* and **104***c* share a common direction of periodicity (e.g., the x-direction) and target cells **104***b* and **104***d* share a common direction of periodicity (e.g., the y-direction). The diffraction grating **118** is designed to correspond to these directions of periodicity. For example, the diffraction grating cells **118***a* and **118***c* correspond to target cells **104***a* and **104***c* and have a periodicity in the x-direction. By way of another example, the diffraction grating cells **118***b* and **118***d* correspond to target cells **104***b* and **104***d* and have a periodicity in the y-direction.
[0083] In embodiments, the diffraction grating **118** may be configured as any type of reflective or transmissive grating. For example, the diffraction grating **118** may be a phase grating.
[0084] It should be understood that the diffraction grating **118** shown in FIG. **3**A is for illustrative purposes only and not intended to be limiting. For example, the diffraction grating **118** may be uniform across the field (e.g., not segmented) so that the target **104** is illuminated by all mutually coherent illumination beams **108**. However, this may not be preferred from a light budget point of view because more light may be required to image the sample **106**. Additionally, the diffraction grating **118** may be segmented even more than shown in FIG. **3**A to correspond to each sub-cell in the target **104**. For example, each diffraction grating cell (e.g., diffraction grating cell **104***a*) may include two or more sub-cells.
[0085] FIG. **4** illustrates a flow diagram illustrating a method **400**, in accordance with one or more embodiments of the present disclosure. Applicant notes that the embodiments and enabling technologies described previously herein in the context of the overlay metrology system **100** should be interpreted to extend to method **400**. It is further noted, however, that the method **400** is not limited to the architecture of the overlay metrology system **100**.
[0086] In embodiments, the method **400** includes a step **402** of generating, with an incoherent illumination source, incoherent illumination.
[0087] A single incoherent illumination source may provide sufficient illumination to perform overlay metrology on a sample.
[0088] In embodiments, the method **400** includes a step **404** of diffracting, with a diffraction grating, the incoherent illumination into one or more pairs of mutually coherent illumination beams.
[0089] For example, the diffraction grating may be designed such that a single pair of mutually coherent beams are formed from the incoherent illumination. By way of another example, the diffraction grating may be designed such that two pairs of mutually coherent beams are formed from the incoherent illumination.
[0090] Additionally, when the periodic features of the metrology target are arranged into two or more target cells, the diffraction grating may include periodic features arranged into two or more diffraction grating cells with a common arrangement as the two or more target cells. The directions of periodicity of features in the two or more diffraction grating cells may match directions of periodicity of the periodic features in the two or more target cells.
[0091] In embodiments, the method **400** includes a step **406** of directing, with an objective lens, the one or more pairs of mutually coherent illumination beams to a metrology target on a sample, wherein the metrology target is configured in accordance with a metrology recipe to include periodic features associated with two or more lithographic exposures.
[0092] In embodiments, the method **400** includes a step **408** of collecting, with the objective lens, sample light associated with diffraction of the one or more pairs of mutually coherent illumination beams. Thus, a single objective lens may be used to direct the one or more pairs of mutually coherent illumination to the sample and direct sample light associated with diffraction of the one or more pairs of mutually coherent illumination beams to a detector.
[0093] In embodiments, the method **400** includes a step **410** of passing, with a mask, a single

nonzero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams.
[0094] In embodiments, the method **400** includes a step **412** of blocking, with a mask, a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams.
[0095] For example, the mask may block all other orders diffraction except for a single +1-order or −1-order diffraction associated with each beam (e.g., allowing a +1-order diffraction associated with a first beam and a −1-order diffraction associated with a second beam). Additionally, other orders of diffraction may be diffracted outside of the collection pupil.
[0096] In embodiments, the method **400** includes a step **414** of generating, with a detector, an image of the metrology target based on light passed by the mask. Because 0-order diffraction is blocked by the mask, a dark field image may be generated for the metrology target.
[0097] In embodiments, the method **400** includes a step **416** of generating one or more metrology measurements of the sample based on the image in accordance with the metrology recipe.
[0098] Metrology measurements may be generated for the sample when the sample has a metrology target with features having periodicity in a single measurement direction. Additionally, metrology measurements may be generated when the metrology target has a first set of periodic features along a first measurement direction and a second set of periodic features along a second measurement direction.
[0099] The method **400** may further include a step of generating correctables for one or more process tools based on the one or more metrology measurements. For example, the correctables based on one or more metrology measurements may be used to control a fabrication tool using any combination of feed-forward or feedback control techniques. As an illustration, feedback control may be used to compensate for deviations of a fabrication tool for various samples within a lot or series of lots. As another illustration, feed-forward control may be used to compensate for deviations measured at one process step for a sample or series of samples when performing a subsequent process step. Any type of fabrication tool may be controlled such as, but not limited to, a lithography tool (e.g., a scanner, a stepper, or the like), an etching tool, or a polishing tool.
[0100] It is contemplated that when the sample has a metrology target with features having periodicity in a single measurement direction, a single pair of mutually coherent illumination beams, wherein the single pair of mutually-coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the single measurement direction may be used to generate metrology measurements for the sample. Further, when the metrology target has a first set of periodic features along a first measurement direction and a second set of periodic features along a second measurement direction, two pairs of mutually coherent illumination beams may be required to fully perform metrology measurements. The two pairs of mutually coherent illumination beams may include a first pair of mutually coherent illumination beams, wherein the first pair of mutually coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the first measurement direction and a second pair of mutually coherent illumination beams, wherein the second pair of mutually-coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the second measurement direction.
[0101] The herein described subject matter sometimes illustrates different components contained within, or connected with, other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “connected” or “coupled” to each other to achieve the desired functionality, and any two

components capable of being so associated can also be viewed as being "couplable" to each other to achieve the desired functionality. Specific examples of couplable include but are not limited to physically interactable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interactable and/or logically interacting components.

[0102] It is believed that the present disclosure and many of its attendant advantages will be understood by the foregoing description, and it will be apparent that various changes may be made in the form, construction, and arrangement of the components without departing from the disclosed subject matter or without sacrificing all of its material advantages. The form described is merely explanatory, and it is the intention of the following claims to encompass and include such changes. Furthermore, it is to be understood that the invention is defined by the appended claims.

Claims

1. A metrology system comprising: an incoherent illumination source to generate incoherent illumination; a diffraction grating to split the incoherent illumination into one or more pairs of mutually coherent illumination beams; an objective lens to direct the one or more pairs of mutually coherent illumination beams to a metrology target on a sample, wherein the metrology target is configured in accordance with a metrology recipe to include periodic features associated with two or more lithographic exposures, wherein the objective lens further collects sample light associated with diffraction of the one or more pairs of mutually-coherent illumination beams; one or more masks, wherein the one or more masks are configured in accordance with the metrology recipe to pass a single nonzero-order diffraction beam and block a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams; a detector configured to generate an image of the metrology target based on light passed by the one or more masks; and a controller communicatively coupled to the detector, wherein the controller includes one or more processors configured to execute program instructions causing the one or more processors to generate one or more metrology measurements of the sample based on the image in accordance with the metrology recipe.

2. The metrology system of claim 1, wherein the incoherent illumination source comprises a lamp source.

3. The metrology system of claim 1, wherein the incoherent illumination source comprises a supercontinuum source.

4. The metrology system of claim 1, wherein the incoherent source comprises a broadband illumination source, wherein at least the objective lens is corrected for chromatic aberration, wherein the image is color-corrected.

5. The metrology system of claim 1, wherein the one or more masks comprise an annular mask with a central aperture.

6. The metrology system of claim 1, wherein the one or more masks comprise an annular mask with a central obscuration.

7. The metrology system of claim 1, wherein the diffraction grating is located at a field plane conjugate to the sample.

8. The metrology system of claim 1, wherein the periodic features of the metrology target have a periodicity along a single measurement direction.

9. The metrology system of claim 8, wherein the one or more pairs of mutually coherent illumination beams comprise a single pair of mutually coherent illumination beams, wherein the single pair of mutually coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the single measurement direction.

10. The metrology system of claim 8, wherein the one or more measurements are associated with the single measurement direction.

11. The metrology system of claim 1, wherein the periodic features of the metrology target comprise: a first set of periodic features along a first measurement direction; and a second set of periodic features along a second measurement direction.
12. The metrology system of claim 11, wherein the one or more pairs of mutually coherent illumination beams comprise: a first pair of mutually coherent illumination beams, wherein the first pair of mutually coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the first measurement direction; and a second pair of mutually coherent illumination beams, wherein the second pair of mutually coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the second measurement direction.
13. The metrology system of claim 1, wherein the periodic features of the metrology target are arranged into two or more target cells, wherein the diffraction grating includes periodic features arranged into two or more diffraction grating cells with a common arrangement as the two or more target cells, wherein directions of periodicity of features in the two or more diffraction grating cells match directions of periodicity of the periodic features in the two or more target cells.
14. The metrology system of claim 1, wherein the diffraction grating comprises a phase grating.
15. The metrology system of claim 1, wherein the mask is a pupil mask at a collection pupil.
16. The metrology system of claim 1, further comprising: a patterned mirror, wherein the patterned mirror is configured to direct the one or more pairs of mutually coherent illumination beams to the sample and pass the single nonzero-order diffraction beam to the detector.
17. The metrology system of claim 16, wherein the patterned mirror is configured to pass a single nonzero-order diffraction beam and block a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams.
18. The metrology system of claim 1, wherein the one or more masks comprises: a patterned mirror.
19. A metrology method comprising: generating, with an incoherent illumination source, incoherent illumination; diffracting, with a diffraction grating, the incoherent illumination into one or more pairs of mutually coherent illumination beams; directing, with an objective lens, the one or more pairs of mutually coherent illumination beams to a metrology target on a sample, wherein the metrology target is configured in accordance with a metrology recipe to include periodic features associated with two or more lithographic exposures; collecting, with the objective lens, sample light associated with diffraction of the one or more pairs of mutually coherent illumination beams; passing, with one or more masks, a single nonzero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams; blocking, with the one or more masks, a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually coherent illumination beams; generating, with a detector, an image of the metrology target based on light passed by the one or more masks; and generating one or more metrology measurements of the sample based on the image in accordance with the metrology recipe.
20. The metrology method of claim 19, wherein generating, with an incoherent illumination source, incoherent illumination comprises: generating, with a lamp source, incoherent illumination.
21. The metrology method of claim 19, wherein generating, with an incoherent illumination source, incoherent illumination comprises: generating, with a supercontinuum source, incoherent illumination.
22. The metrology method of claim 19, wherein generating, with an incoherent illumination source, incoherent illumination comprises: generating, with a broadband source, incoherent illumination, wherein at least the objective lens is corrected for chromatic aberration, wherein the image is color-corrected.
23. The metrology method of claim 19, wherein the periodic features of the metrology target have a

periodicity along a single measurement direction.

24. The metrology method of claim 23, wherein the one or more pairs of mutually coherent illumination beams comprise a single pair of mutually coherent illumination beams, wherein the single pair of mutually coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the single measurement direction.

25. The metrology method of claim 19, wherein the periodic features of the metrology target comprise: a first set of periodic features along a first measurement direction; and a second set of periodic features along a second measurement direction.

26. The metrology method of claim 25, wherein the one or more pairs of mutually coherent illumination beams comprise: a first pair of mutually coherent illumination beams, wherein the first pair of mutually coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the first measurement direction; and a second pair of mutually coherent illumination beams, wherein the second pair of mutually coherent illumination beams is incident on the metrology target at opposing azimuth angles aligned with the first measurement direction.

27. The metrology method of claim 19, wherein the periodic features of the metrology target are arranged into two or more target cells, wherein the diffraction grating includes periodic features arranged into two or more diffraction grating cells with a common arrangement as the two or more target cells, wherein directions of periodicity of features in the two or more diffraction grating cells match directions of periodicity of the periodic features in the two or more target cells.

28. A metrology system comprising: a diffraction grating comprising one or more diffraction grating cells having a direction of periodicity configured to split incoherent illumination from an incoherent illumination source into one or more pairs of mutually coherent illumination beams; an objective lens to direct the one or more pairs of mutually coherent illumination beams to a metrology target on a sample, wherein the metrology target is configured in accordance with a metrology recipe including two or more lithographic exposures to generate periodic features having a direction of periodicity arranged into one or more target cells, wherein the direction of periodicity of a target cell corresponds to the direction of periodicity of a diffraction grating cell, wherein the objective lens further collects sample light associated with diffraction of the one or more pairs of mutually coherent illumination beams; one or more masks, wherein the one or more masks are configured in accordance with the metrology recipe to pass a single nonzero-order diffraction beam and block a zero-order diffraction beam associated with each of the mutually coherent illumination beams of the one or more pairs of mutually-coherent illumination beams; a detector configured to generate an image of the metrology target based on light passed by the one or more masks; and a controller communicatively coupled to the detector, wherein the controller includes one or more processors configured to execute program instructions causing the one or more processors to generate one or more metrology measurements of the sample based on the image in accordance with the metrology recipe.